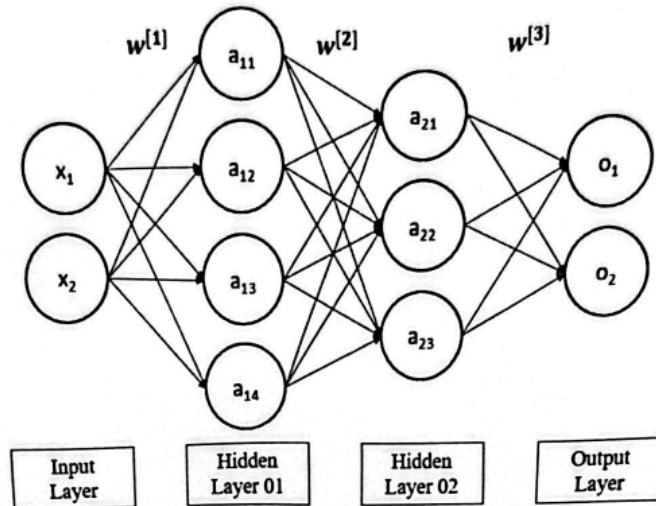


Problem 02 Back Propagation – 21 marks

Consider the neural network diagram given below:



- $w_{ij}^{[k]}$ represents the value of the weight belonging to the k^{th} layer (shown in the diagram above), between the i^{th} node in the input layer and j^{th} node in the output layer. For example, the weight between the nodes a_{12} and a_{23} will be represented as $w_{23}^{[2]}$.

$w_{11}^{[1]}$	$w_{12}^{[1]}$	$w_{13}^{[1]}$	$w_{14}^{[1]}$	$w_{21}^{[1]}$	$w_{22}^{[1]}$	$w_{23}^{[1]}$	$w_{24}^{[1]}$
0.9	0.75	-0.15	0.33	-0.79	-0.03	0.1	0.02
$w_{11}^{[2]}$	$w_{12}^{[2]}$	$w_{13}^{[2]}$	$w_{21}^{[2]}$	$w_{22}^{[2]}$	$w_{23}^{[2]}$	$w_{31}^{[2]}$	$w_{32}^{[2]}$
-0.58	1.0	-0.8	0.23	0.45	0.05	0.25	-0.65
$w_{33}^{[2]}$	$w_{41}^{[2]}$	$w_{42}^{[2]}$	$w_{43}^{[2]}$	$w_{11}^{[3]}$	$w_{12}^{[3]}$	$w_{21}^{[3]}$	$w_{22}^{[3]}$
-0.09	-0.57	0.51	-0.33	0.97	0.3	-0.01	-0.2
$w_{31}^{[3]}$	$w_{32}^{[3]}$	_____	_____	_____	_____	_____	_____
0.85	0.76	_____	_____	_____	_____	_____	_____

- The activation function for the hidden layers is *tanh* and for the output layer it is *sigmoid*.
- The cost function for the neural network is *binary cross entropy*.
- The learning rate used for weight updates is *0.01*.
- The input values for x_1 and x_2 are *0.5* and *0.3* respectively.
- The actual values (ground truth) for o_1 and o_2 are *0.4* and *0.45* respectively.
- Assume that the biases are *0*.

Part A – Forward Pass (4+3+2 marks)

1. Using the appropriate weights and activation function, compute the values of the nodes for the **first hidden layer**.

$a_{11}: w_{11}x_1 + w_{21}x_2$ $(0.9)(0.5) + (-0.79)(0.3)$	$z_{a_{11}}^{(1)} = 0.213, O_{a_{11}} = 0.2098$
$a_{12}: w_{12}x_1 + w_{22}x_2$ $(0.75)(0.5) + (-0.03)(0.3)$	$z_{a_{12}}^{(1)} = 0.366, O_{a_{12}} = 0.3504$
$a_{13}: w_{13}x_1 + w_{23}x_2$	$z_{a_{13}} = -0.045, O_{a_{13}} = -0.0449$
$a_{14}: w_{14}x_1 + w_{24}x_2$ $(0.33)(0.5) + (0.02)(0.3)$	$z_{a_{14}}^{(1)} = 0.171, O_{a_{14}} = 0.1693$

2. Using the appropriate weights and activation function, compute the values of the nodes for the **second hidden layer**.

$$a_{21}: w_{11}^{(2)} O_{a_{11}} + w_{21}^{(2)} O_{a_{12}} + w_{31}^{(2)} O_{a_{13}} + w_{41}^{(2)} O_{a_{14}}$$

$$(-0.58)(0.2098) + (0.23)(0.3504) + (0.25)(-0.0449) + (-0.57)(0.1693)$$

$$z_{a_{21}} = -0.1488, O_{a_{21}} = -0.1477$$

$$a_{22}: w_{12} O_{a_{11}} + w_{22} O_{a_{12}} + w_{32} O_{a_{13}} + w_{42} O_{a_{14}}$$

$$(1.0)(0.2098) + (0.45)(0.3504) + (-0.65)(-0.0449) + (0.51)(0.1693)$$

$$z_{a_{22}} = 0.4830$$

$$O_{a_{22}} = 0.4486$$

$$a_{23}: w_{13} O_{a_{11}} + w_{23} O_{a_{12}} + w_{33} O_{a_{13}} + w_{43} O_{a_{14}}$$

$$(-0.8)(0.2098) + (0.05)(0.3504) + (-0.09)(-0.0449) + (-0.33)(0.1693)$$

$$z_{a_{23}} = -0.2021, O_{a_{23}} = +0.2071 - 0.1993$$

3. Using the appropriate weights and activation function, compute the values of the nodes for the **output layer**.

$$\begin{aligned}
 O_1: & w_{11}^{(3)} O_{a21} + w_{21}^{(3)} O_{a22} + w_{31}^{(3)} O_{a23} \\
 \textcircled{1} \quad Z_{01} &= (0.97)(-0.1477) + (-0.01)(0.4486) + (0.85)(-0.1993) \\
 Z_{01} &= -0.3171, \quad O_{01} = \textcircled{0.4213} \\
 O_2: & w_{12}^{(3)} O_{a21} + w_{22}^{(3)} O_{a22} + w_{32}^{(3)} O_{a23} \\
 & (0.3)(-0.1477) + (0.2)(0.4486) + (0.76)(-0.1993) \\
 Z_{02} &= -0.2854, \quad O_{02} = 0.4291
 \end{aligned}$$

Part B – Backward Propagation and Weight Update (3+3+3+3 marks)

1. By using partial derivatives, compute the derivative of the error w.r.t. the weights $w_{31}^{(3)}$.
 (Hint: You may find it helpful to use the derivatives you computed in Problem 1)

$$\begin{aligned}
 \frac{\partial L}{\partial w_{31}^{(3)}} &= \frac{\partial E_1}{\partial O_{01}} \cdot \frac{\partial O_{01}}{\partial Z_{01}} \cdot \frac{\partial Z_{01}}{\partial w_{31}^{(3)}} \\
 E &= -y \log(\hat{y}) - (1-y) \log(1-\hat{y}) \\
 \frac{\partial E}{\partial \hat{y}} &= \frac{-y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \\
 &= -0.949 + 1.036 \\
 &= 0.087
 \end{aligned}$$

$$\frac{\partial O_{01}}{\partial Z_{01}} = (1 - O_{01}) O_{01} = (1 - 0.4213)(0.4213) = 0.2438$$

$$\frac{\partial Z_{01}}{\partial w_{31}^{(3)}} = -0.1993$$

$$\frac{\partial L}{\partial w_{31}^{(3)}} = 0.087 \times 0.2438 \times -0.1993 = -0.00422$$

- (Hint: You may find it helpful to use the derivatives you computed in the previous part)

$$= 0.0007$$

- (Hint: You may find it helpful to use the derivatives you computed in the previous part)

$$= (0.087)(-0.1993)(0.85)(-0.1595)(0.05)(0.8773)(0.5) +$$

$$(-0.0853)(0.2449)(0.76)(-0.1595)(0.05)(0.8773)(0.5) +$$

$$(0.087)(-0.1993)(-0.01)(0.7988)(0.45)(0.8773)(0.5) +$$

$$(-0.853)(0.2449)(-0.2)(0.7988)(0.45)(0.8773)(0.5) +$$

$$\begin{aligned}
 & (0.087)(-0.1993)(0.97)(0.9782)(0.23)(0.8773)(0.5) + \\
 & (-0.853)(0.2449)(0.3)(0.9782)(0.23)(0.8773)(0.5) \\
 & = 0.00005 + 0.00005 + 0.00002 + 0.0065 + \\
 & (-0.0016) + (-0.0061)
 \end{aligned}$$

$$= -0.0010$$

4. Use the computed derivatives to update the weights $w_{31}^{[3]}$, $w_{23}^{[2]}$ and $w_{12}^{[1]}$.

$$w_{31}^{(3)} = w_{31}^{(3)} - \alpha \frac{\partial E}{\partial w_{31}^{(3)}} = 0.85 - (0.01)(-0.00422)$$

$$= 0.8500422$$

$$w_{23}^{(2)} = 0.05 - (0.01)(0.0007) = 0.0499$$

$$w_{12}^{(1)} = 0.75 - (0.01)(-0.0010)$$

$$w_{12}^{(1)} = 0.75001$$